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October 19, 1992
SUPERSEDING
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April 9, 1973

FEDERAL SPECIFICATION

TOOL BOX, PORTABLE; GENERAL SPECIFICATION FOR

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specifications covers the general requirements for portable tool boxes used for carrying and storing hand tools. Classification and detail requirements for a particular tool box are covered by the applicable detail specification.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-P-40	- Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Federal Standards:

FED-STD-187	- Identification of Pressed Bends, Forms, Seams, and Joints, (Sheet Metal)
FED-STD-376	- Preferred Metric Units for General use by the Federal Government

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Detail specifications. The individual requirements shall be as specified herein and in accordance with the applicable detail specifications.

3.2 Material. The materials shall be as specified. Materials not definitely specified shall be of the quality best suited for the purpose intended.

3.2.1 Sheet metal. The sheet metal from which the tool boxes are fabricated shall be commercial quality cold rolled sheet steel. The thickness of the sheet steel gage specified shall apply after fabrication of the box and shall conform to table I.

TABLE I. Sheet steel thickness (inches)

Gage number	Nominal thickness	Tolerance (plus or minus)
14	0.0747	0.005
16	0.0598	0.005
18	0.0478	0.004
20	0.0359	0.003
22	0.0299	0.003
24	0.0239	0.003
26	0.0179	0.002

3.2.2 Plywood. The plywood used in the construction of tool boxes shall be equal to or exceed the requirements for commercial grade AD interior type.

3.2.3 Pressed bends, forms, seams, and joints (sheet metal). The identification of pressed bends, forms, seams, and joints shall be in accordance with FED-STD-187.

3.3 Construction. Construction details shall be as shown in the applicable detail specifications and shall be capable of passing the specified tests.

3.3.1 Seams. Seams shall be spot welded, riveted, continuous welded, or lock seamed.

3.3.2 Welding. All welding shall be accomplished by the resistance welding method (spot welding). All welds shall be in accordance with good welding practices, and of a size and spacing adequate for the tool box to pass the load, drop, and weld tests (see 4.3.1, 4.3.2, and 4.3.4). Spot welds shall be stronger than the base metal surrounding the weld area (see 4.3.4).

3.3.3 Exposed edges. All exposed edges shall be free from burrs and sharp edges.

3.3.4 Rivets. Rivets shall be made of steel, properly applied, and adequate for the application. Shouldered (stepped shank) rivets shall be used for tray lever pivots except when the tray lever pivots on a dimpled bearing surface. Blind rivets shall not be used in cantilever tray pivot points. Blind rivets used in other applications shall be of the filled core type.

3.3.5 Tolerances. Unless otherwise specified, the outside dimensional tolerances shall be ± 1 inch; out of parallel or out of square tolerance shall be ± 1/8 inch.

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3.3.6 Hinges. The lids and bodies of all boxes shall be joined by a continuous (piano type) hinge having a length approximating the overall length of the tool box. The hinge shall be of a suitable size for the purpose intended and be formed as an integral part of the box and lid, or be welded to the box body and lid. The hinge pins used in the hinge shall be made of steel, brass, or bronze and shall be not less than 0.080 inch in diameter. All hinge pins including those used in hasp hinges shall be made to be nonremovable except by means which would cause distortion or damage to the tool box.

3.3.7 Hardware. Hardware (hasps, staples, integral locks, drawbolts, and catches) is intended to hold the tool box in a closed or locked position. All hardware shall be securely attached to the box. The thickness of the sheet metal in the hardware shall be not less than 20 gage. When wire is used for loops in drawbolts, it shall be not less than 1/8 inch diameter. Hasps, staples, and similar hardware shall accommodate a 1-3/4 inch padlock and be so designed that, with the padlock in the locked position, the box cannot be entered without damage or distortion to the box. Unless otherwise specified, integral locks shall be of the pin or disk tumbler type which are operated by a grooved key. Two fitted keys shall be furnished with each lock.

3.3.8 Handle. All tool boxes shall be provided with a handle or handles, as specified in the detail specification, suitable for carrying the box. The handle(s) shall be welded or riveted to the box. When riveted, a back-up plate (doubler) or other adequate reinforcement shall be provided to prevent the rivets from loosening or pulling through.

3.4 Coating. Box bodies, lids, trays, and drawers, excluding hardware, shall have a durable red baked finish in accordance with good commercial practice. Hardware shall be finished either the same as the box bodies or plated as furnished commercially.

3.5 Dimensions. Dimensions not specified shall be in accordance with good commercial practice and within other limiting dimensions and capacities. The overall dimensions of tool boxes, excluding projections, shall be width (dimension from end to end), depth (dimension from front to back), and height (dimension from surface to the bottom of the box).

3.6 Marking. Each tool box shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture. The marking shall be as durable as the coating on the box.

3.7 Workmanship. Workmanship shall be in accordance with high grade industrial practices. Lids shall be attached to the box bodies in a manner that will permit their easy opening and closing. Trays shall lift in and out without binding, catches and hasps shall be in proper alignment, and handles and corner reinforcements shall be securely attached.

3.8 Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this Federal Specification are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105. Sampling for drop tests and hinge pin tests shall be as specified in 4.3.2 and 4.3.3.

4.2.1 Visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in terms of tool boxes. The sample unit shall be one tool box. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

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TABLE II. End item visual defects

Examine	Defect
Construction and Workmanship	Not as specified
	Seams not welded, riveted, or seamed as required
	Not correct welding method
	Size and spacing of welds inadequate
	Spot welds not stronger than base metal
	Burrs or sharp edges
	Incorrect or improperly applied rivet
	Hinge incorrect size or type
	Back-up plate missing from riveted handles
	Lids do not open and close easily
	Trays bind or catch when lifted out
	Catches or hasps not properly aligned
	Handle(s) or corner reinforcements not securely attached
Materials	Not as specified
	Sheet metal not as specified (see 3.2.1)
	Plywood does not meet requirements for commercial grade AD interior type
Finish	Not as specified
	Not specified color
	Not smooth, uniform
Marking	Missing, incomplete, or not legible

4.2.2 Dimensional examination. The end items shall be examined for any nonconformance to dimensional requirements. The lot size shall be expressed in terms of tool boxes. The sample unit shall be one tool box. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.3 Test methods. Unless otherwise specified in the applicable detail specification, the tool boxes shall be tested in accordance with 4.3.1, 4.3.2, 4.3.3, and 4.3.4. The lot size shall be expressed in units of tool boxes. The sample unit shall be one tool box. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.3.1 Load test. Each sample unit shall be loaded as specified in the applicable detail specification with tools, lead shot, or other bulk material. The box shall then be closed and lifted by its handle(s) 25 times to a height of not less than 12 inches from the floor to the bottom of the tool box. The box shall be rejected if any permanent distortion or damage results which would affect its serviceability.

4.3.2 Drop test. The sample unit shall be loaded as specified in table III-A (in lieu of loadings specified in the applicable detail specification) with tools, lead shot, or other bulk material. The box shall then be flatly dropped 6 times in the normal upright position, from a height shown in table III-B, onto a concrete surface. The box shall be rejected if the hinges do not operate properly; if the lid cannot be opened and closed properly; if any rivets or welds have pulled loose; if any seams have pulled open; or if any brackets, levers, trays, or drawers are distorted so that they are not functional for the purpose intended.

TABLE III-A. Loading rates of tool box areas for drop tests

Area of box (determine square feet using inside dimensions)	Loading rate in pounds per square feet
Cantilevered tray, drawer, or compartment less than 3 inches deep	10
Drawer or compartment (other than box bottom) over 3 inches deep and any tote tray	15
Box bottom	20

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TABLE III-B. Height of free fall drop for tool boxes

Weight of loaded tool box	Height of free fall drop
0 to 15 pounds	27 inches
15 to 30 pounds	22 inches
30 to 50 pounds	19 inches
50 to 75 pounds	17 inches
75 to 110 pounds	15 inches
110 to 150 pounds	14 inches
Over 150 pounds	12 inches

4.3.3 Hinge pin test. The sample unit shall be tested to determine if the hinge pin can be removed from any of the exterior hinges or hasps on the box without distorting the box. The box shall be rejected if any hinge pin can be removed without obvious damage to the box.

4.3.4 Weld test (applicable to boxes with spot welds). One sample box shall be tested to determine the integrity of the spot welds. At least ten spot welds shall be selected in random locations and tested by driving a chisel between the welded components in an unwelded area, or by peeling one sheet back against the weld until failure occurs in the weld or in the base metal around the periphery of the weld. If failure occurs in a welded area, the spot welding shall be considered unsatisfactory and the tool box considered defective.

4.4 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table IV. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE IV. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.1).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Levels of preservation, packaging, and packing required (see 5.1).
c. Types and sizes, etc. (see applicable detail specification).

MILITARY INTERESTS:

Military Coordinating Activity

Army - GL

Custodians

Army - GL
Air Force - 99

Review Activity

Air Force - 84

User Activities

Army - AT, ME
Navy - MC, YD

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

GSA - FSS

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